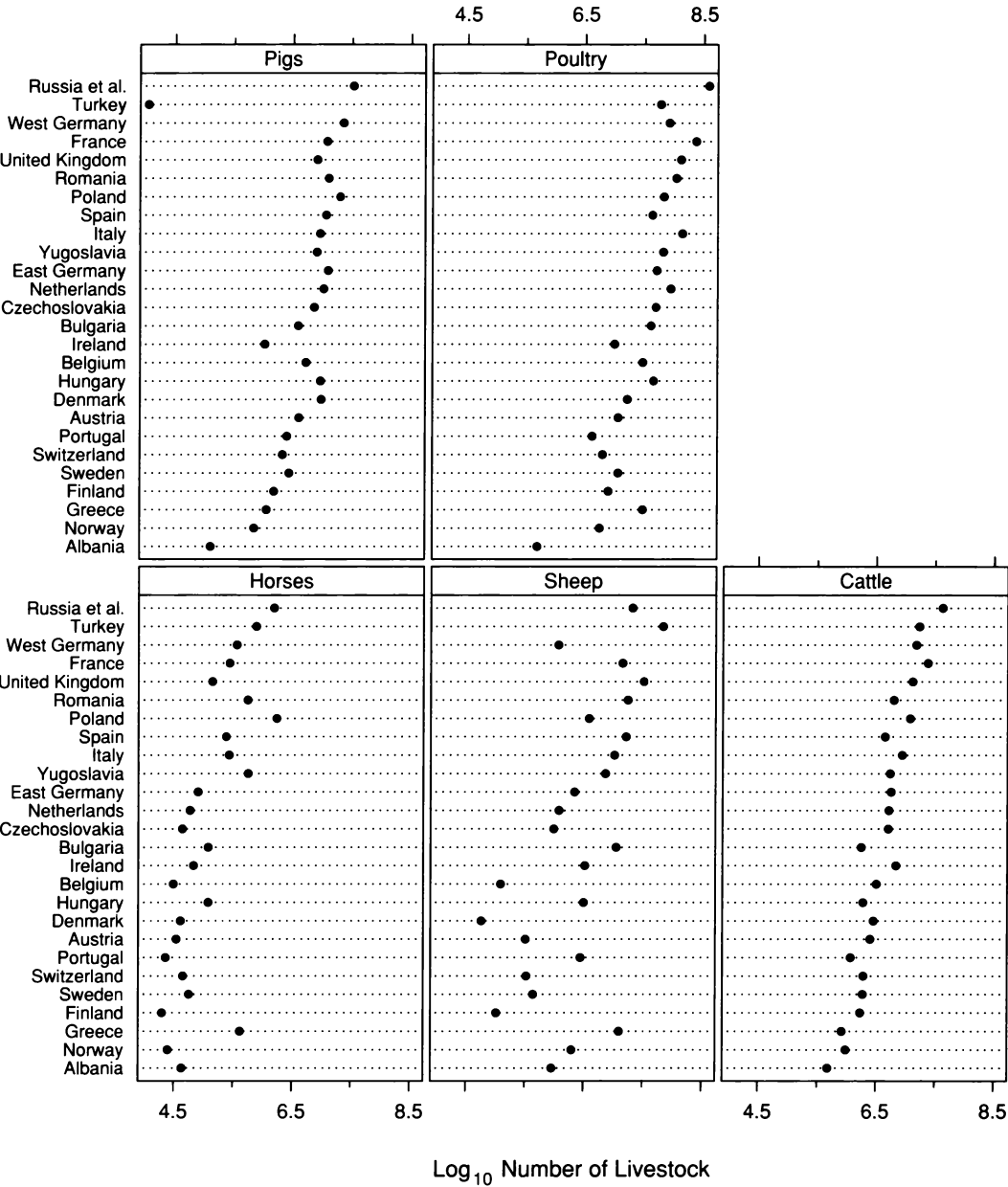




6.1 A multiway dot plot displays multiway data: log counts of five types of livestock in 26 countries.

6 *Multiway Data*

Figure 6.1 graphs the logarithms of livestock counts from a 1987 census of farm animals in 26 countries [14]. Logs are taken at the outset because the counts vary by about 4.5 powers of 10; if the data were graphed on the original scale, only the variability in a few large observations would be visible. Russia *et al.* is the European part of Russia and the European countries that were formerly part of the Soviet Union.

The census has a purpose that would be hard to guess — to study air pollution. The feces and urine of livestock produce ammonia, an air pollutant. The livestock counts are combined with estimates of the amounts of ammonia emitted by the wastes of different animals to provide a release rate of ammonia into the atmosphere of Europe. The counts in Figure 6.1 lead to an estimate of 5.2 megatons per year; this accounts for 81% of all ammonia emissions.

The log counts in Figure 6.1 are *multiway data*. There is one quantitative variable, log count, and two categorical variables, country and livestock type. The quantitative variable is a response, and the goal is to study how it depends on the categorical variables, which are factors. What distinguishes multiway data is the cross-classification of the categorical variables; there is a value of the response for each combination of levels of the two categorical variables. In this case, there is one log count for each combination of country and livestock type.

6.1 *Multiway Dot Plots*

The visualization method of Figure 6.1 is a *multiway dot plot*. There are *panels*, the individual dot plots of the display, and there are *levels*, the rows of each panel. In Figure 6.1, the livestock variable is encoded by the panels and the country variable is encoded by the levels.

The countries are assigned to the levels so that the country medians increase from bottom to top. The median of the five observations for Albania is the smallest country median. Norway has the next smallest country median, and Russia *et al.* has the largest. The panels are ordered so that the livestock medians increase in graphical order: from left to right and from bottom to top. The median for horses is the smallest, and the median for poultry is the largest.

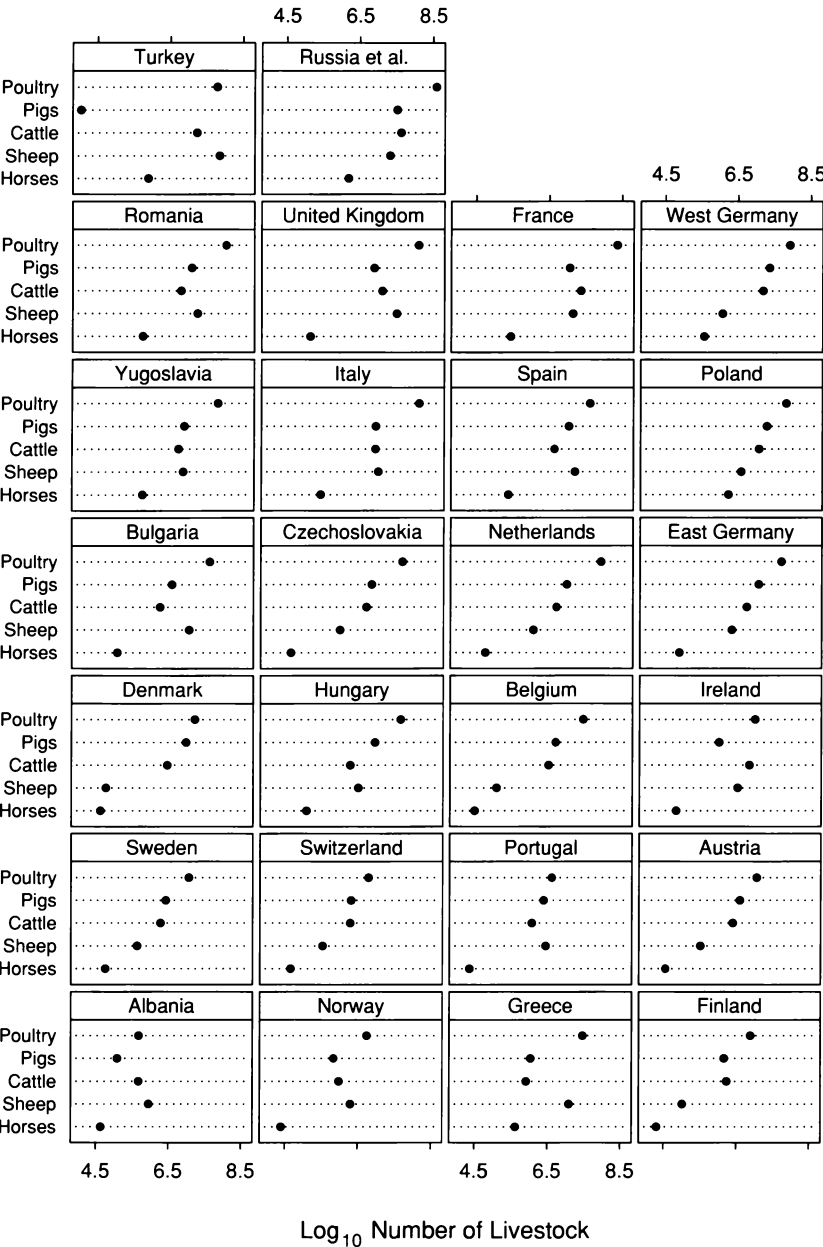
Figure 6.1 shows that the amount of variation in the log counts for sheep is greater than that of the other livestock types; in other words, the ordering of the sheep data agrees the least well with the ordering of the country medians. The log pig counts show interesting behavior. There is an unusually low number of pigs in Turkey. Even though it is near the top in log counts overall, it has the fewest pigs. Albania has a small number of pigs as well, but its value is not nearly as deviant as that of Turkey because overall, Albania is at the bottom in log counts. And Ireland also has a small number of pigs in the sense that other countries nearby in the median ordering have more pigs.

Figure 6.2 displays the data by a second multiway dot plot, but this time the livestock variable has been assigned to the levels, and the country variable has been assigned to the panels. The display shows that for Turkey and Albania, sheep are the most abundant animal; this makes up for the lack of pigs. But for all other countries, poultry are the most abundant. The log count for pigs in Turkey is so small that it is exceeded by the log count of horses, whereas in all other countries, horses are the least abundant category.

Visual Perception

As with all display methods in this book, the display method of the multiway dot plot is based on a careful consideration of visual perception.

One visual issue is an asymmetry in the perception of the effects of the categorical variables. We can more effectively compare values within a panel than values between panels. Consider Figure 6.1 where the levels encode the country and the panels encode the livestock type.

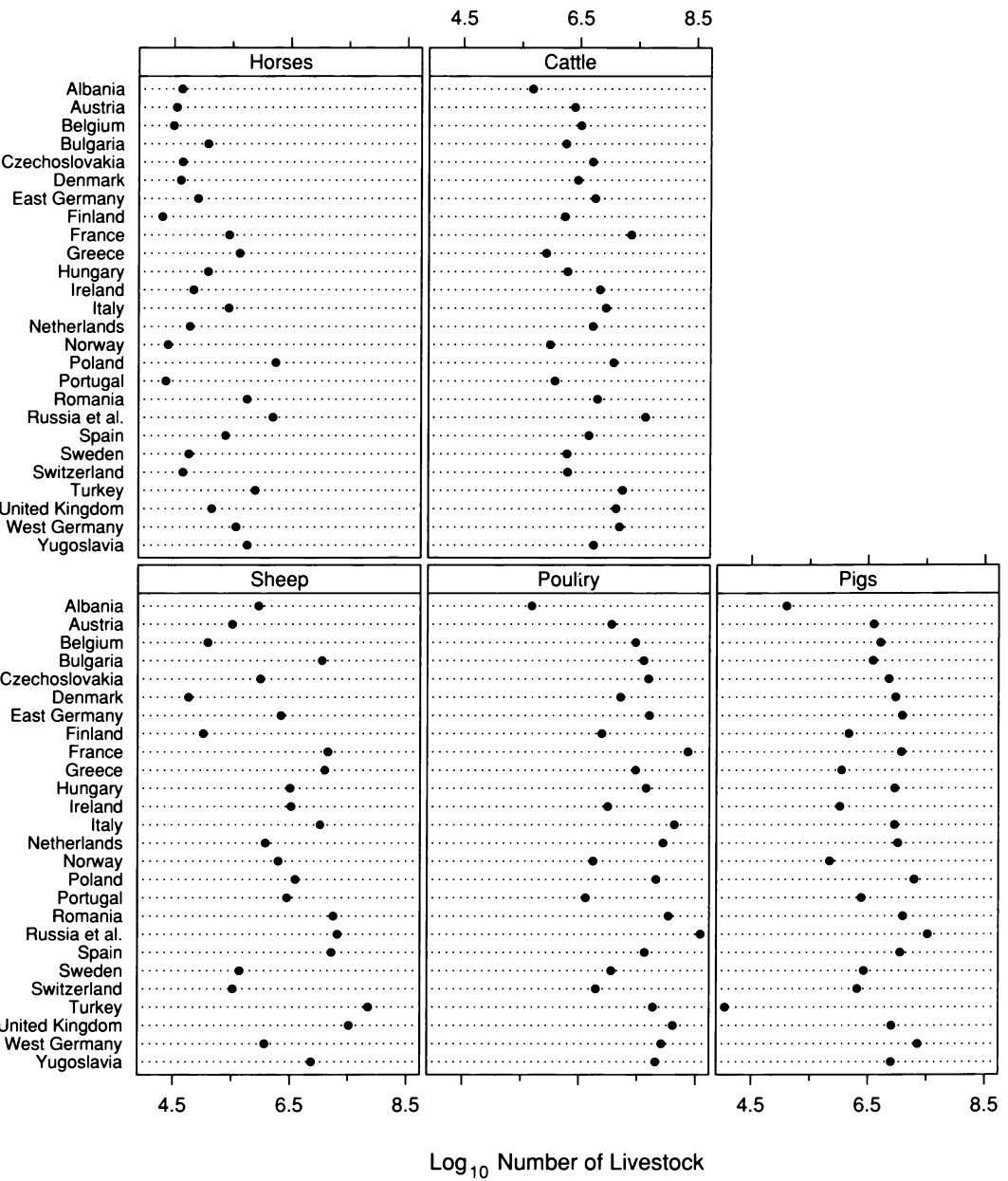


6.2 The log counts are graphed again with the livestock variable assigned to the levels and the country variable assigned to the panels.

The country values for each livestock type can be perceived as a gestalt because they are graphed along a single scale line. For example, this contributes to our ability to perceive the greater variation in the sheep data. We can, of course, readily compare these gestalts from one panel to the next; this allows us to assess how the distribution of values changes as the panel category changes. But in Figure 6.1 we cannot compare the five values for a given country as effectively. This is done far better in Figure 6.2, where the livestock type is assigned to the levels. For example, we can readily perceive that in Finland, poultry have the largest log count and horses have the smallest; this cannot be as readily perceived in Figure 6.1. Because of this asymmetry, it is often important to explore multiway data by as many multiway dot plots as there are categorical variables, with each variable assigned once to the levels.

Another visual issue is the orderings of the levels and panels by a measure of the category locations, in this case, the median. The orderings are crucial to the perception of effects. The ordering of the levels by country median in Figure 6.1 establishes gestalts on each panel that are easier to compare from one panel to the next. For example, this allows us to see that the sheep behave differently from the other livestock types. Also, the level ordering provides a benchmark for the log count for each livestock type — the values of the nearby log counts. For example, Figure 6.1 shows that the small log cattle count in Albania is not unusually small given the overall rank of Albania, but the number of pigs in Ireland is unusually small given the overall rank of Ireland.

In Figure 6.3, the log counts are displayed again with the levels ordered alphabetically and the panels ordered arbitrarily. Many of the effects readily seen in Figure 6.1 are not revealed. The sheep data no longer stand out as particularly unusual, and we cannot see that the log pig count in Ireland is low or that the log sheep count in Greece is high.



6.3 The levels and panels on this multiway dot plot of the log counts are not ordered by category median. The levels are ordered alphabetically and the panels are ordered arbitrarily.

6.2 Additive Fits

There are many mathematical functions that can be fitted to multiway data [52]. The simplest is an additive function.

An Additive Parametric Function

Let $y_{\ell c}$ denote the log count for the ℓ th livestock category and the c th country category. The additive function for the log livestock counts is a parametric function of ℓ and c ,

$$\mu + \alpha_{\ell} + \beta_c .$$

The parameter μ measures the overall value of the response. We will constrain the remaining parameters, which are the *main effects*, by

$$\sum_{\ell=1}^5 \alpha_{\ell} = \sum_{c=1}^{26} \beta_c = 0 .$$

The parameters α_{ℓ} are the livestock main effects, and the parameters β_c are the country main effects.

The additive function does not provide for an interaction between country and livestock. The change in the function in going from livestock type i to livestock type j , $\alpha_j - \alpha_i$, is the same for all countries. Similarly, the change in the function in going from country i to country j , $\beta_j - \beta_i$, is the same for all livestock types. If the additive function provided a good fit to the data, the underlying patterns of the log counts on the panels of either Figure 6.1 or 6.2 would have the same overall shape but simply shift left and right. There would be no interaction between country and livestock type. This is clearly not the case for the log livestock counts; for example, the sheep pattern is quite different from the others. But even when an interaction is present in multiway data, fitting an additive function can be useful; the interaction becomes part of the residuals, and visualization of the residuals can give a clearer picture of the properties of the interaction than we get in exploratory graphs of the unfitted data.

Least-Squares and Bisquare

Least-squares can be used to fit an additive function to multiway data. For the log livestock counts, $y_{\ell c}$, least-squares determines values of μ , α_ℓ , and β_c that minimize

$$\sum_{\ell=1}^5 \sum_{c=1}^{26} (y_{\ell c} - \mu - \alpha_\ell - \beta_c)^2.$$

The minimizing values, which are the estimates of the parameters, have a simple form. The estimate of μ is the overall mean of the data,

$$\hat{\mu} = \frac{1}{130} \sum_{\ell=1}^5 \sum_{c=1}^{26} y_{\ell c}.$$

The estimate of α_ℓ is the mean across countries for livestock type ℓ , minus the overall mean,

$$\hat{\alpha}_\ell = \frac{1}{26} \sum_{c=1}^{26} y_{\ell c} - \hat{\mu}.$$

The estimate of β_c is the mean across livestock types for country c , minus the overall mean,

$$\hat{\beta}_c = \frac{1}{5} \sum_{\ell=1}^5 y_{\ell c} - \hat{\mu}.$$

The fitted values are

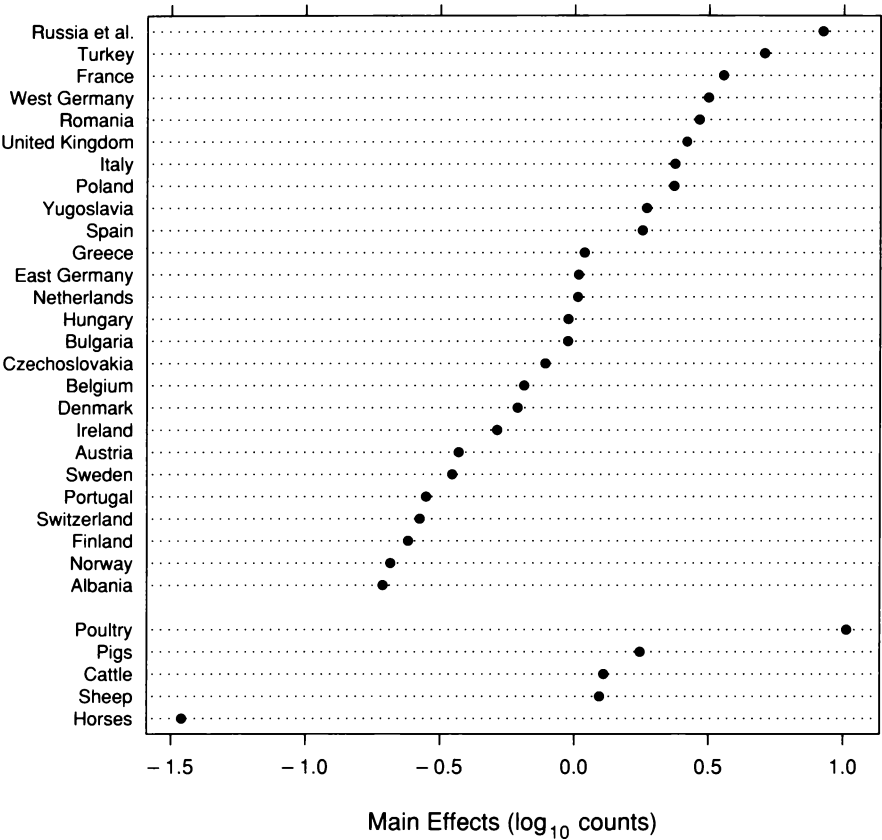
$$\hat{y}_{\ell c} = \hat{\mu} + \hat{\alpha}_\ell + \hat{\beta}_c$$

and the residuals are

$$\hat{\varepsilon}_{\ell c} = y_{\ell c} - \hat{y}_{\ell c}.$$

If the initial visualization of multiway data by dot plots suggests aberrant behavior in the data, bisquare fitting can be used to fit the additive function. The robustness iterations are the same as those for the data types discussed in earlier chapters. First, the function is fitted by least-squares; in succeeding steps, robustness weights are computed and the function is re-fitted using weighted least-squares. Quite clearly, the log livestock counts contain aberrant behavior, for example, the exceptionally low log pig count in Turkey, so we will use bisquare.

Figure 6.4 graphs the bisquare estimates of the main effects. Russia *et al.* have the largest country main effect, and Albania has the smallest. Poultry have the largest livestock main effect and horses have the smallest. It seems likely that the country main effects are related to the populations and areas of the countries, but we will not check this further here. The country main effects vary by about 1.5 on the log scale, which is a factor of $10^{1.5} \approx 30$. The livestock main effects vary by about 2.5, which is a factor of 10 greater. The country main effects are spread out uniformly across their range, but the livestock main effects have one large value, one small value, and three similar values somewhat above the middle of the range.



6.4 The dot plot displays the bisquare estimates of the country and livestock main effects.

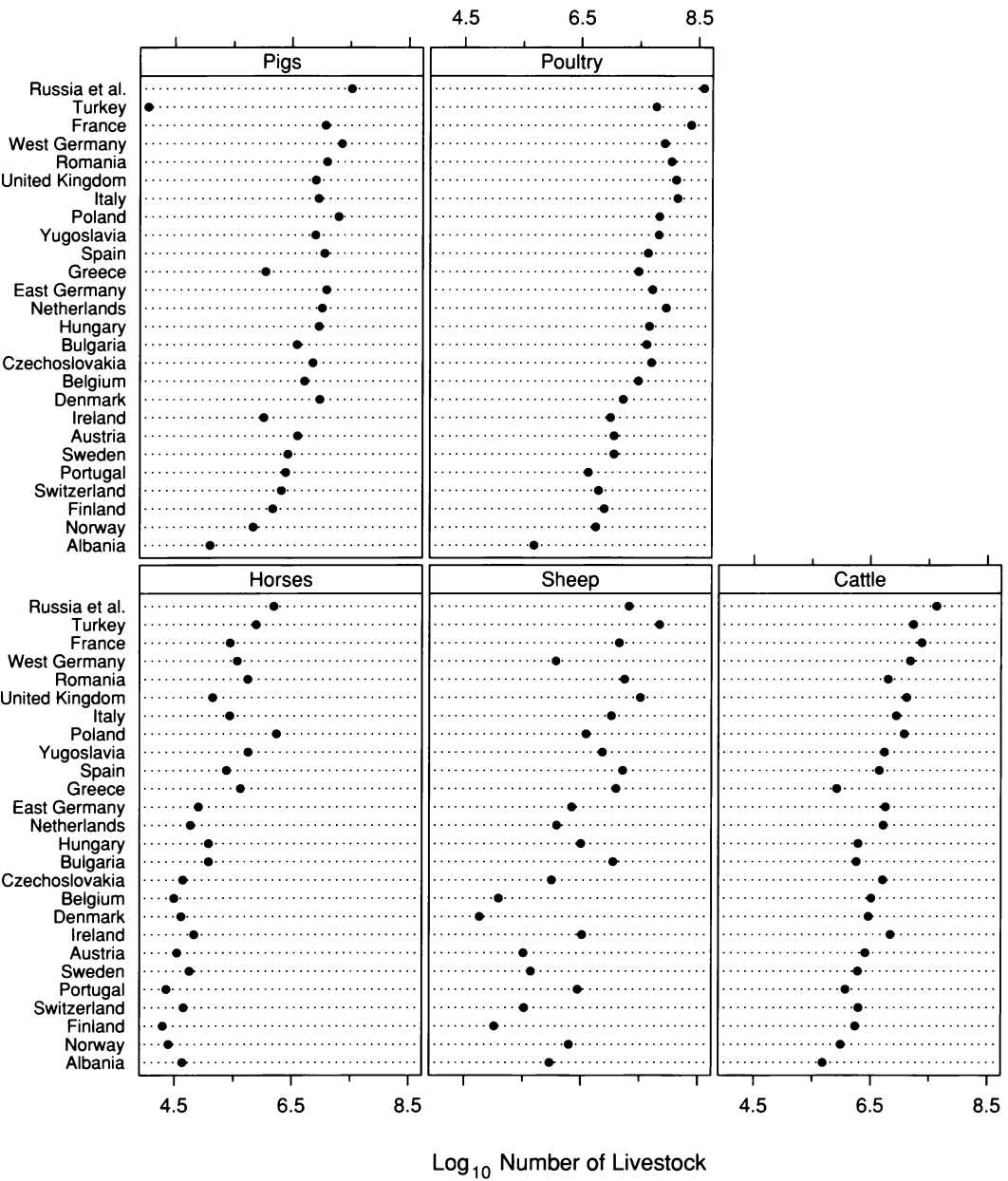
The display method of the multiway dot plot, as discussed in detail earlier, orders the levels and panels by category medians. For the log livestock counts, each country median is a location measure of the five log counts for the country, and each livestock median is a location measure of the 26 log counts for the livestock type. The bisquare fitting has provided new location measures. The livestock locations are

$$\hat{\mu} + \hat{\alpha}_{\ell}$$

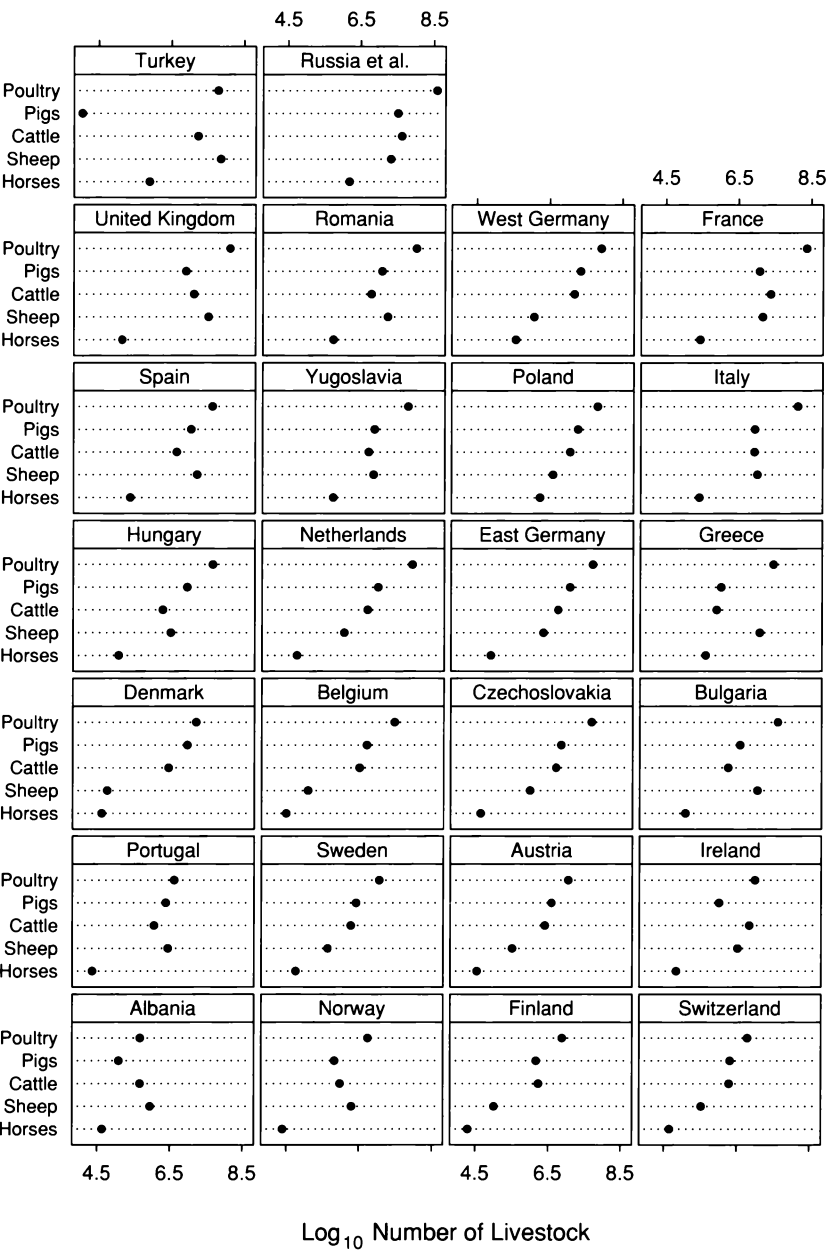
and the country locations are

$$\hat{\mu} + \hat{\beta}_c.$$

Figures 6.5 and 6.6 are dot plots with the panels and levels ordered by the new location measures, which, of course, is the same as ordering by the estimates of the main effects. No substantially new insight is gained, in part because the new and old orderings are similar. But the new ordering will be used for subsequent visualization of the residuals and fitted values, and having Figures 6.5 and 6.6 will be important when we are stimulated by patterns in the subsequent graphs to look back at the data.



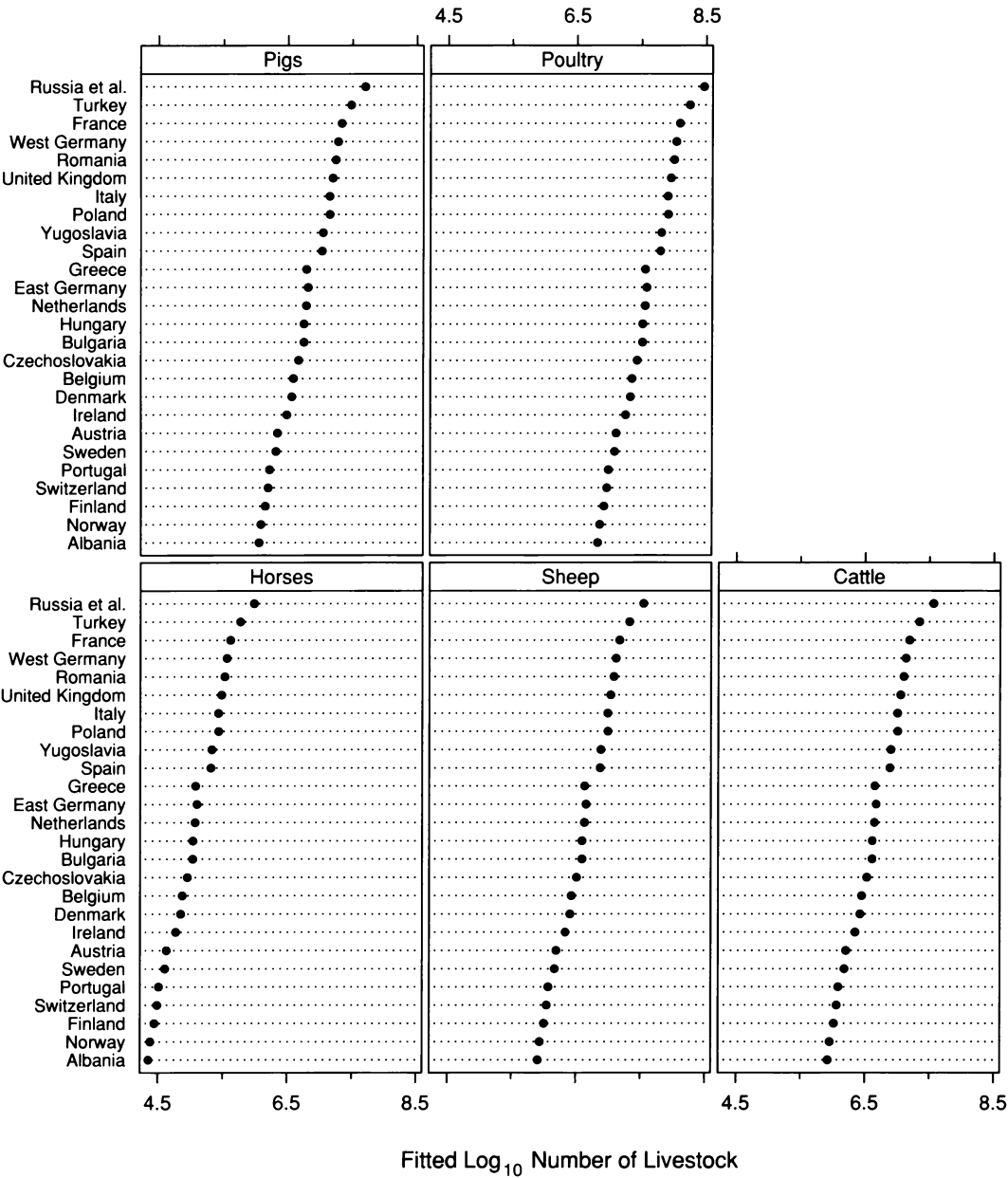
6.5 On this multiway dot plot of the log counts, the levels and panels are ordered by the bisquare estimates of the main effects.



6.6 On this multiway dot plot of the log counts, the levels and panels are ordered by the bisquare estimates of the main effects.

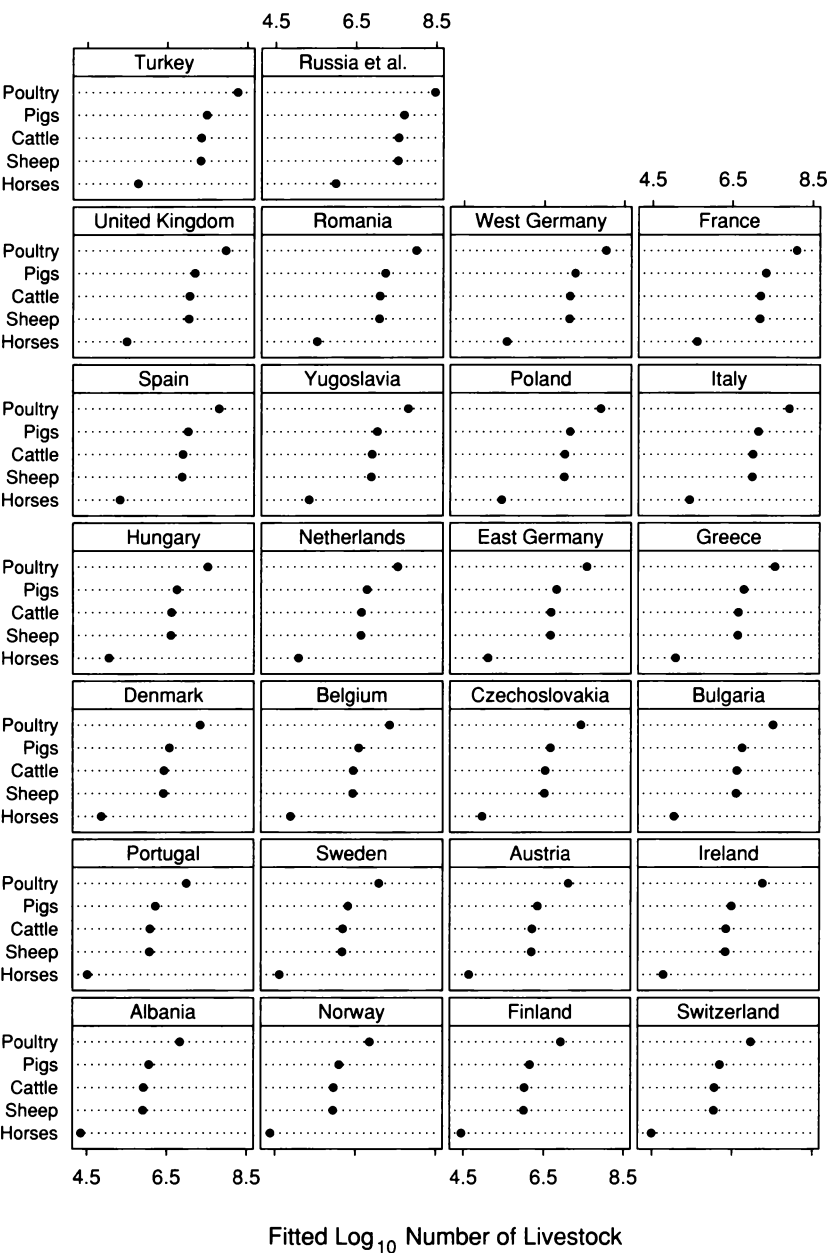
Graphing Fitted Values and Residuals

Figures 6.7 and 6.8 graph the fitted values of the log counts. For each dot plot, the shapes of the patterns of the points on the panels are the



6.7 The multiway dot plot shows the additive fit to the log counts.

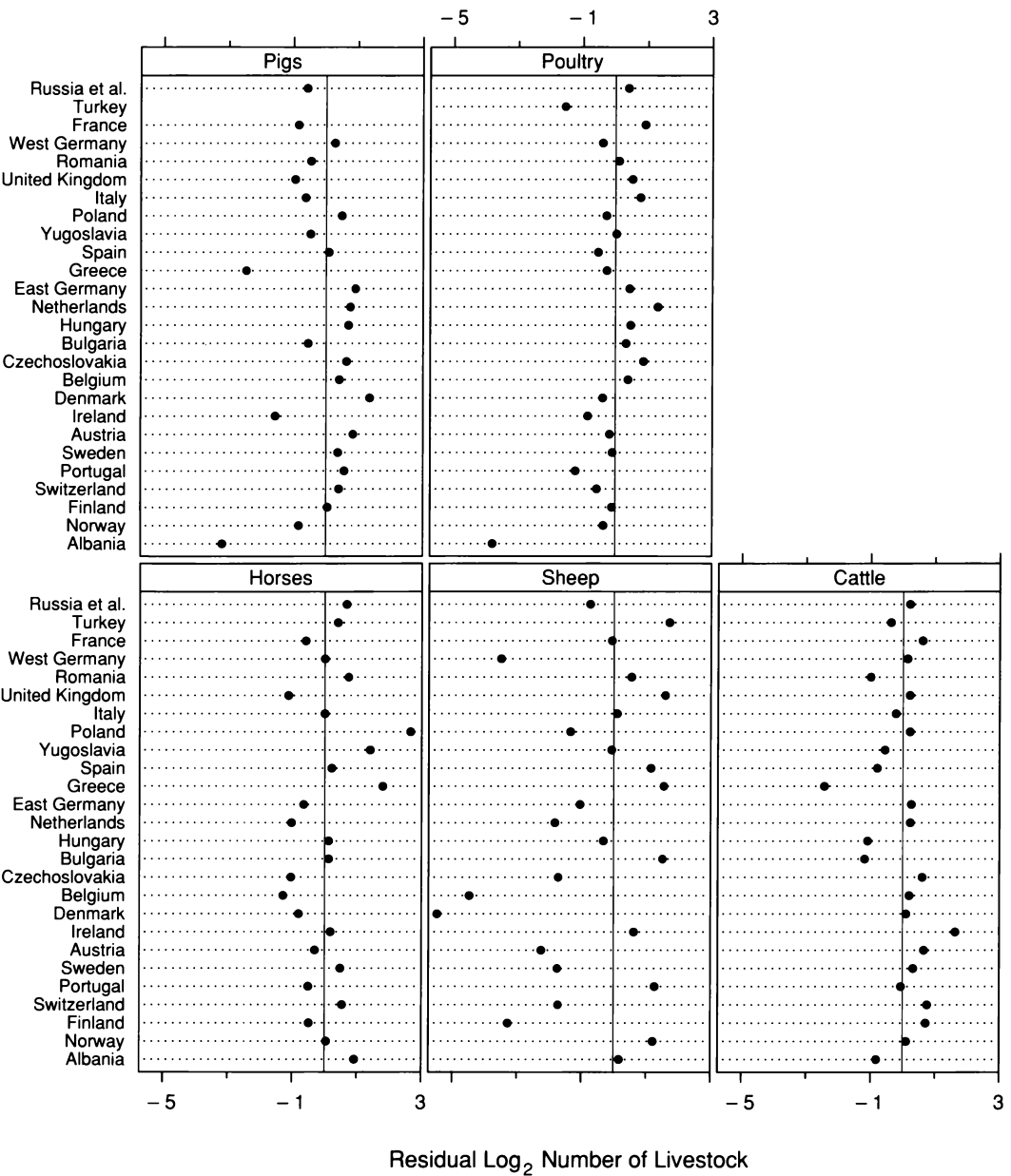
same, and the patterns simply shift left and right according to the main effects. As explained earlier, this is a feature of the additivity of the fitted function, which results in no interaction between the two categorical variables.



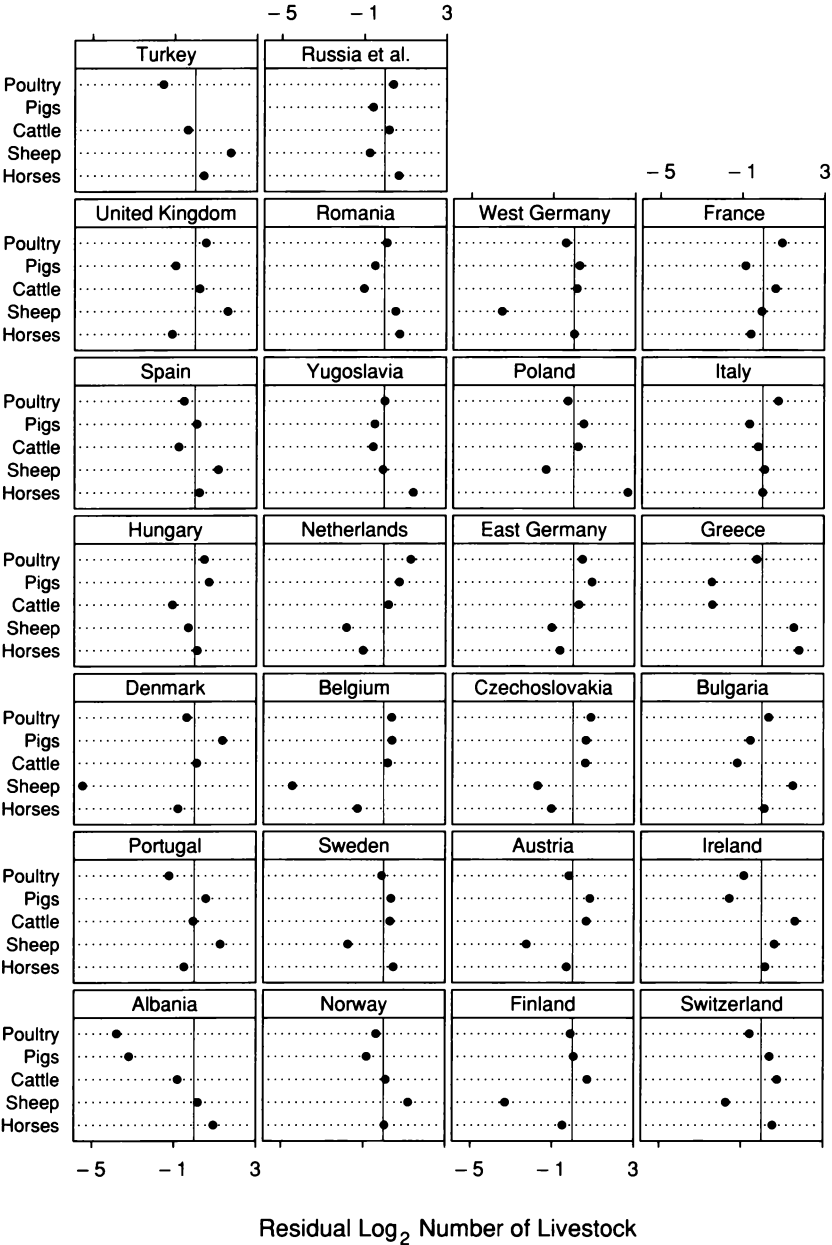
6.8 The multiway dot plot shows the additive fit to the log counts.

The residuals are graphed in Figures 6.9 and 6.10. The scale of the residuals is log base 2 because most of them vary only by about 1 on a log base 10 scale. The residual for pigs in Turkey has been omitted because it ruins the resolution of the display; including it requires an expansion of the horizontal scale by about 70%.

A large residual indicates an over-population of the livestock type in the country, and a small residual indicates an under-population. For example, Poland has the largest horse residual, $2.7 \log_2$ count. Given the main effect for Poland and the main effect for horses, there are more horses in Poland than one would expect by a factor of $2^{2.7} = 6.5$. In other words, Poland is horse country. Turkey and Albania have the smallest pig residuals. For both countries, the under-population of pigs is due to religious dietary laws. Turkey and Greece have the greatest over-population of sheep; anyone who has been to a Turkish or Greek restaurant knows why. Denmark has the greatest under-population of sheep, but perhaps to make up for it, the greatest over-population of pigs.



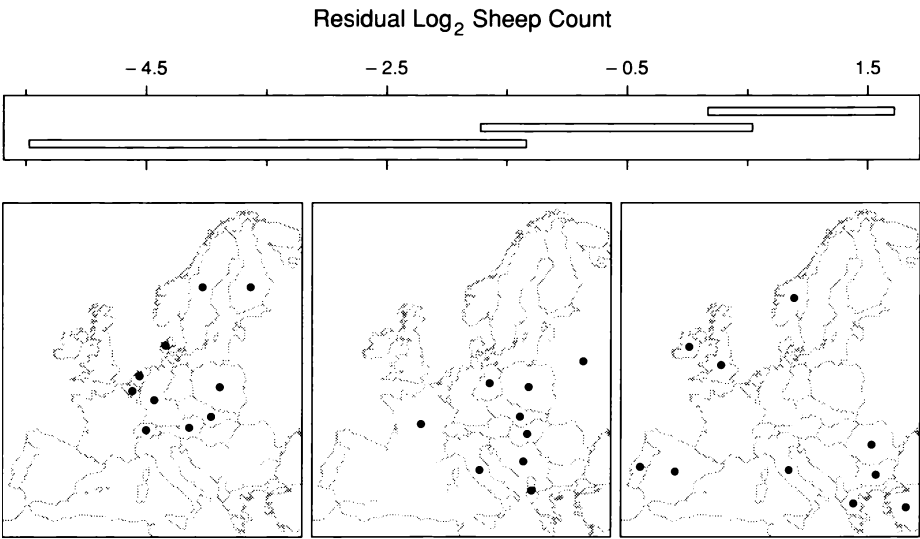
6.9 The multiway dot plot displays the residuals from the additive fit to the log counts.



6.10 The multiway dot plot displays the residuals from the additive fit to the log counts.

The residuals are clearly not homogeneous. Sheep have greater residual variation than the other livestock types. For many countries, the sheep residual is either the largest or the smallest of the five residuals for the country. Careful study of the multiway dot plots of the residuals suggests a geographic influence. This is investigated in Figure 6.11; the residuals are displayed by a level plot. The equal-count method has been used to select the three intervals, whose fraction of overlap is 0.25.

Geography does appear to account for a part of the sheep variation. Countries in a contiguous region covering central Europe and most of Scandinavia have the smallest residuals; that is, they have fewer sheep than predicted by the fitted function. As we move outward from this region, the residuals tend to increase and are largest at the boundaries of the region to the south and northwest. We could fit the dependence on geographical location with loess, but we will not do so, since we have treated the data sufficiently for our purposes.



6.11 The level plot shows the dependence of the sheep residuals on geographical location.

6.3 *Superposition and Differences*

Sorting data from smallest to largest is one of the fundamental tasks of software systems. Through the decades, as computers developed, so did sorting algorithms, with ever increasing advances in efficiency. By 1992, one might have thought that the practical limits in efficiency had been nearly reached, and that any further work would yield only small gains.

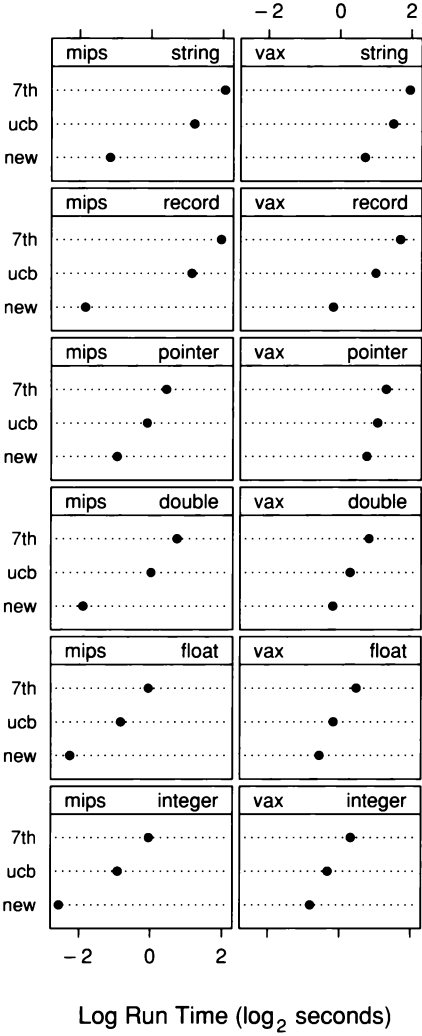
But in 1992, Jon Bentley and Douglas McIllroy became interested in *qsort*, a sort program in UNIX[®], because of reports of disastrously poor performance for several highly structured examples. For example, *qsort* was utterly undone by the triangular sequence

$$1, 2, \dots, n-1, n, n, n-1, \dots, 2, 1.$$

Bentley and McIllroy promptly invented a new algorithm that solved the problems. And while they were at it, they set new speed records for general sequences, beating the old ones by a wide margin.

As part of their investigation of run-time performance, Bentley and McIllroy generated random sequences of length 10^4 , and sorted them with three algorithms: their new one, the *qsort* program in the Berkeley version of UNIX, and the *qsort* program in the Seventh Edition version. They tested on two machines: a VAX 8550 and a MIPS R3000. There were six input types for the sequences: integer, float, double, pointer, record, and string. For each combination of algorithm, machine, and input type they generated and sorted 10 sequences and computed the average run time. The times are multiway data with $3 \times 2 \times 6 = 36$ observations. It makes sense to take logs at the outset since multiplicative factors are important, not absolute run times.

Figure 6.12 graphs the log run times. The algorithm factor has been assigned to the panel levels. The panels are arranged in a rectangular array with the machine factor assigned to the columns and the input factor assigned to the rows. The ordering of the levels and panels depends on the category medians as it did for the exploratory dot plots of the livestock data. The algorithm medians increase as the levels increase from bottom to top, the input medians increase from bottom to top through the rows of the panels, and the machine medians increase from left to right through the columns.



6.12 The multiway dot plot graphs log run times for three algorithms, two machines, and six input types.

Superposition

In Figure 6.12, the assignment of the algorithm factor to the panel levels allows us to readily assess its effect on log run time. The pattern of the three log run times on each panel can be readily perceived, and the 12 patterns can be compared. To enhance the assessment of the effect of